



Department of Chemistry

Environmental Chemistry (2 credit)

CY1010

July 2026

Green chemistry & Industrial Ecology

Course Objectives

Green Chemistry in education is for instruction in the chemical sciences and is designed to cover new trends in incorporating green or sustainable chemistry concepts into the chemistry curriculum and laboratory.

Chemistry

Pros

- Pharmaceuticals that have improved health and extended life
- Fertilizers that have greatly increased food productivity
- Semiconductors that have made possible computers and other modern electronic devices

Cons

- Pollutants
- Toxic substances
- Nonbiodegradable plastic containers.

These have resulted in harm to the environment

The Pollution Prevention Act (PPA) states:

- Pollution should be prevented or reduced at the source whenever feasible
- Pollution that cannot be prevented or reduced should be recycled
- Pollution that cannot be prevented or reduced or recycled should be treated, and
- Disposal or other releases into the environment should be employed only as a last resort

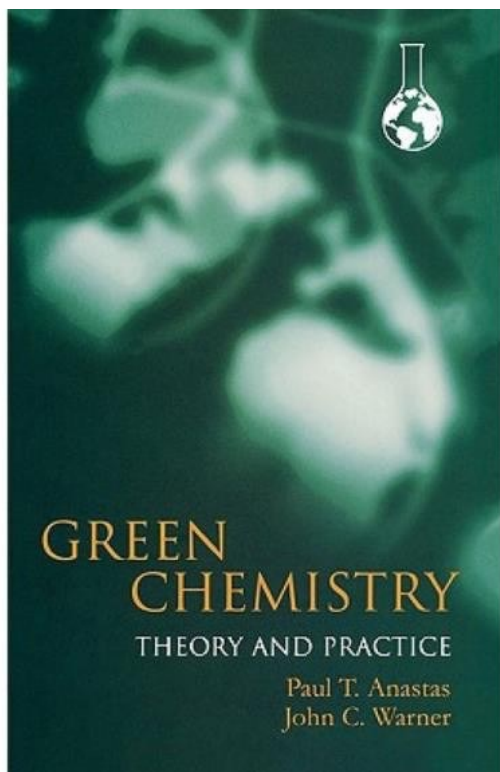
Green Chemistry

- **Green chemistry looks at pollution prevention on the molecular scale**

an extremely important area of chemistry due to the importance of Chemistry in our world today and the implications it can show on our environment.

- **The Green Chemistry program supports the invention of more environmentally friendly chemical processes which reduce or even eliminate the generation of hazardous substances.**
- This program works very closely with the twelve principles of Green Chemistry.

THE 12 PRINCIPLES OF GREEN CHEMISTRY



Anastas, P. T.; Warner, J.C. Green Chemistry: Theory and Practice, Oxford University Press, 1998

Green Chemistry

DEFINITION: Green Chemistry is the utilisation of a set of principles that reduces or eliminates the use or generation of hazardous substances in the design, manufacture and application of chemical products .

GREEN CHEMISTRY IS ABOUT

- Waste Minimisation at Source
- Use of Catalysts in place of Reagents
- Using Non-Toxic Reagents
- Use of Renewable Resources
- Improved Atom Efficiency
- Use of Solvent Free or Recyclable Environmentally Benign Solvent system

12 Principles of Green Chemistry

1. **Prevention.** It is better to prevent waste than to treat or clean up waste after it is formed.
2. **Atom Economy.** Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.
3. **Less Hazardous Chemical Synthesis.** Whenever practicable, synthetic methodologies should be designed to use and generate substances that possess little or no toxicity to human health and the environment.
4. **Designing Safer Chemicals.** Chemical products should be designed to preserve efficacy of the function while reducing toxicity.
5. **Safer Solvents and Auxiliaries.** The use of auxiliary substances (solvents, separation agents, etc.) should be made unnecessary whenever possible and, when used, innocuous.
6. **Design for Energy Efficiency.** Energy requirements should be recognized for their environmental and economic impacts and should be minimized. Synthetic methods should be conducted at ambient temperature and pressure

7. Use of Renewable Feedstocks. A raw material or feedstock should be renewable rather than depleting whenever technically and economically practical.

8. Reduce Derivatives. Unnecessary derivatization (blocking group, protection/deprotection, temporary modification of physical/chemical processes) should be avoided whenever possible .

9. Catalysis. Catalytic reagents (as selective as possible) are superior to stoichiometric reagents.

10. Design for Degradation. Chemical products should be designed so that at the end of their function they do not persist in the environment and instead break down into innocuous degradation products.

11. Real-time Analysis for Pollution Prevention. Analytical methodologies need to be further developed to allow for real-time in-process monitoring and control prior to the formation of hazardous substances.

12. Inherently Safer Chemistry for Accident Prevention. Substance and the form of a substance used in a chemical process should be chosen so as to minimize the potential for chemical accidents, including releases, explosions, and fires.

12 Principles of Green Chemistry

☐ Toxicity

- Reducing the hazard

☐ Feedstocks

- Use of renewable resources

☐ Designing safer products

- Non toxic products by design

☐ Biodegradability

- Enhancing break down at the end of life

☐ Energy

- Reducing the energy needs

☐ Accidents

- Eliminating accidents

☐ Efficiency

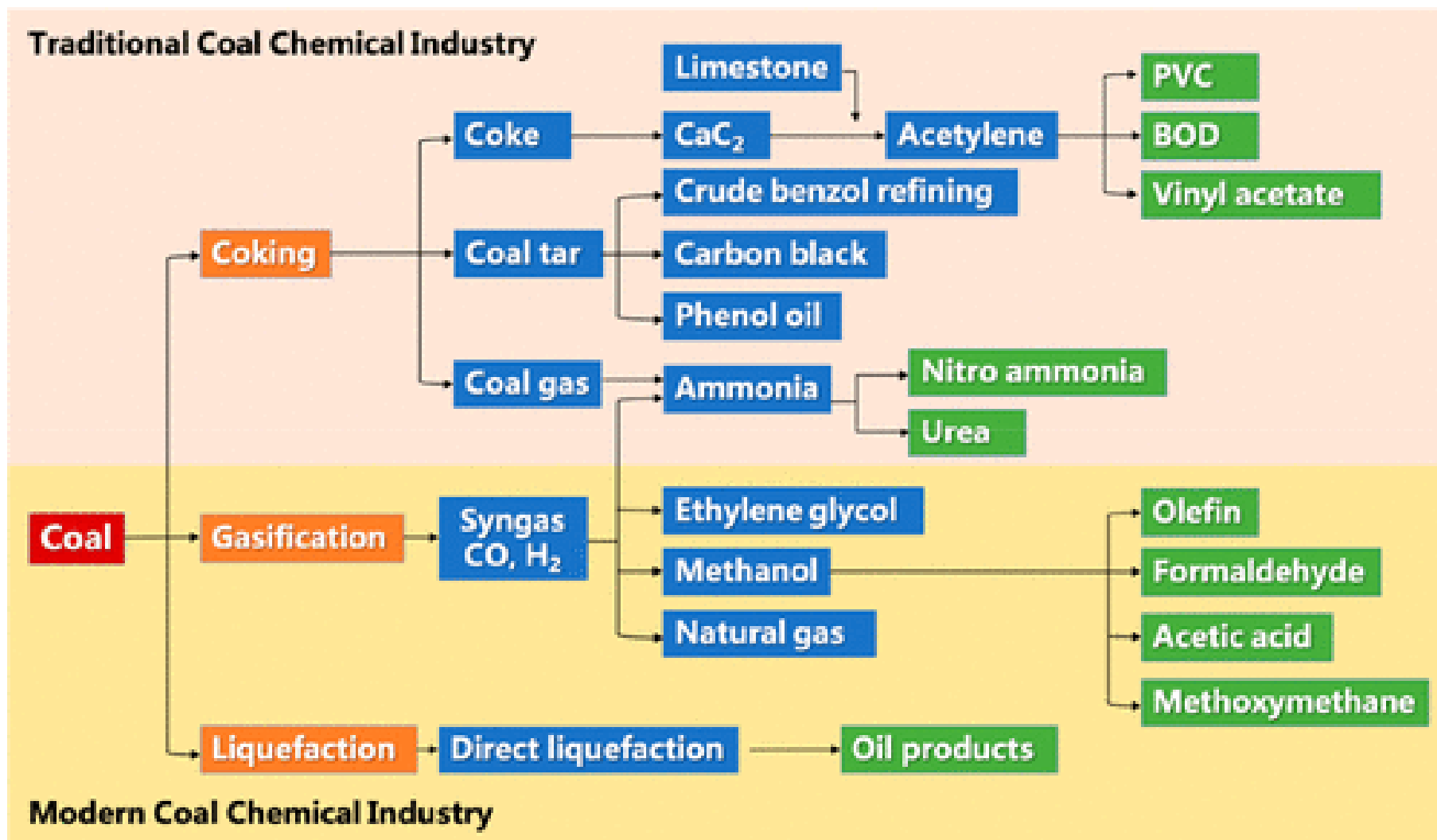
- Shorter processes and synthesis

Feedstocks

- Organic feedstock refers to any raw material that comes from a once-living organism used for its organic carbon and carbon compounds.
- This can include plants fermented for biofuels, recycled cooking oil, animal fats rendered for fuel, and plant material

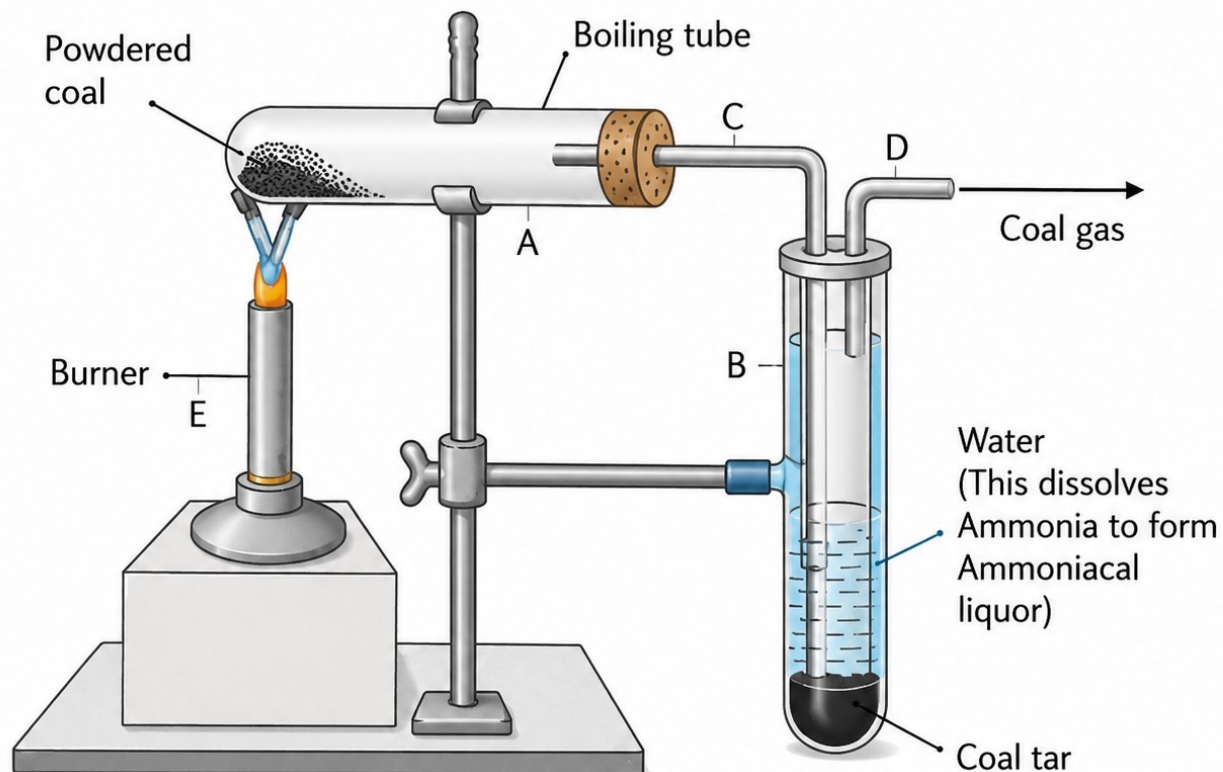


Feedstocks



Destructive distillation of coal

On strong heating of coal in a closed tube it breaks down to different components such as coal gas, coal tar, ammoniacal liquor and coke. This process is called **destructive distillation**.

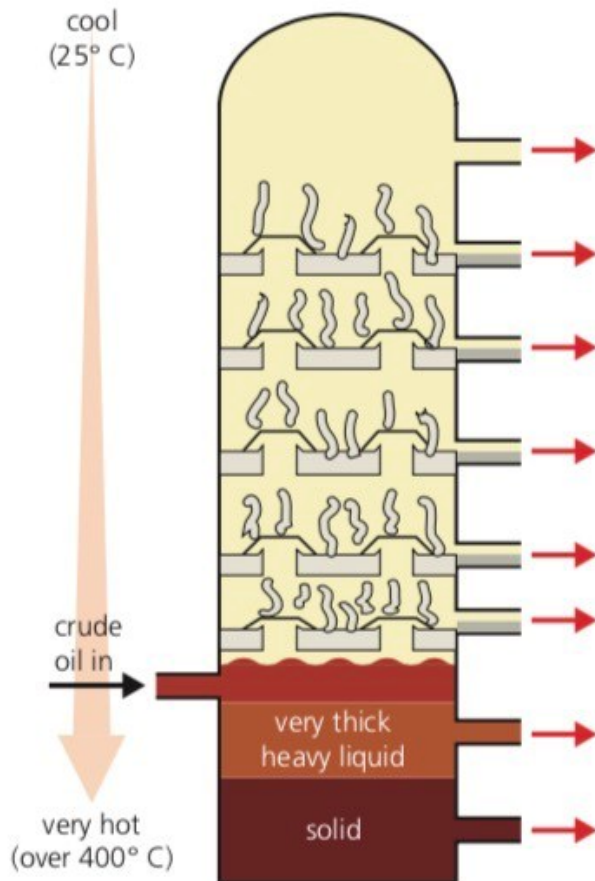


Typical composition of coal tar distillate.

B. P. Limit (°C)	Fraction	Wt%	Main constituents
180	Light oil	up to 3	BTX-aromatics, pyridine bases
210	Carbolic oil	up to 3	Phenols
230	Naphthalene oil	10 – 12	Naphthalene
290	Wash oil	7 – 8	Methylnaphthalenes, acenaphthene
400	Anthracene oil	20 – 28	Anthracene, phenanthrene, carbazole
>400	Pitch	50 – 55	

Feedstocks

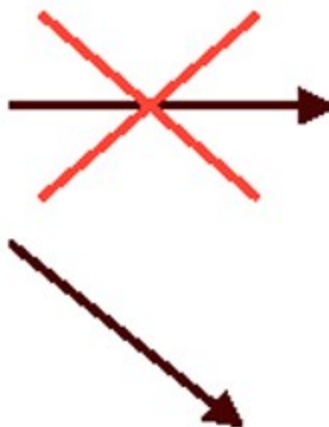
These are the fractions collected after distilling petroleum



Name of fraction	Number of carbon atoms	What fraction is used for	
refinery gas	C ₁ to C ₄	bottled gases for cooking and heating	 boiling points and viscosity increase
gasoline (petrol)	C ₅ to C ₆	fuel for cars	
naphtha	C ₆ to C ₁₀	starting point or feedstock for many chemicals and plastics	
paraffin (kerosene)	C ₁₀ to C ₁₅	fuel for aircraft, oil stoves, and lamps	
diesel oil (gas oil)	C ₁₅ to C ₂₀	fuel for diesel engines	
fuel oil	C ₂₀ to C ₃₀	fuel for power stations, ships, and for home heating systems	
lubricating fraction	C ₃₀ to C ₅₀	oil for car engines and machinery; waxes and polishes	
bitumen	C ₅₀ upwards	for road surfaces and roofs	

Prevention. It is better to prevent waste than to treat or clean up waste after it is formed

**Chemical
Process**



No waste

Waste Prevention

It is better to prevent waste than to treat or clean up waste after it is formed.



Waste Prevention

What is waste?

Waste is anything that is a leftover from the product/process and cannot be re-used.

Waste can be produced in any stage of a process life-cycle:

- Origins of feedstock
- Manufacturing
- Distribution
- Use
- End of life

Waste Prevention

When is waste not waste?

- When it can be reused, becoming a feedstock.
- When it can be reduced or completely eliminated.

Waste Prevention

Waste generated by different industry segments:

Industry segment	Annual product tonnage	Kg waste/Kg product
Oil refining	10^6 – 10^8	ca. 0.1
Bulk chemicals	10^4 – 10^6	<1–5
Fine chemicals	10^2 – 10^4	5–>50
Pharmaceuticals	10 – 10^3	25–>100

R. A. Sheldon, *Chem. Ind.*, 1992, 903–906

R. A. Sheldon, *Chem. Ind.*, 1997, 12–15

Waste Prevention

Many industry sectors try to decrease waste generation by decreasing the number of synthetic steps to generate the product and by implementing a solvent recovery system.

Production of pharmaceuticals produces a lot of waste due making complex molecules with demanding quality standards, often requiring batch processing; there are greater regulatory constraints and often high value and low volume products are manufactured.

R. A. Sheldon, *Chem. Ind.*, 1992, 903–906

R. A. Sheldon, *Chem. Ind.*, 1997, 12–15

Physical Hazards

- Explosives
- Flammable Gases
- Flammable Aerosols
- Oxidizing Gases
- Gases Under Pressure
- Flammable Liquids
- Flammable Solids
- Self-Reactive Substances
- Pyrophoric Liquids
- Pyrophoric Solids
- Self-Heating Substances
- Substances which, in contact with water emit flammable gases
- Oxidizing Liquids
- Oxidizing Solids
- Organic Peroxides
- Corrosive to Metals



Health Hazards

- ☐ Acute Toxicity
- ☐ Skin Corrosion/Irritation
- ☐ Serious Eye Damage/Eye Irritation
- ☐ Respiratory or Skin Sensitization
- ☐ Germ Cell Mutagenicity
- ☐ Carcinogenicity
- ☐ Reproductive Toxicology
- ☐ Target Organ Systemic Toxicity - Single Exposure
- ☐ Target Organ Systemic Toxicity - Repeated Exposure
- ☐ Aspiration Toxicity

Environmental Hazards

- ☐ Hazardous to the Aquatic Environment
- ☐ Acute aquatic toxicity
- ☐ Chronic aquatic toxicity
- ☐ Bioaccumulation potential
- ☐ Rapid degradability

Hazard Communication

Once a chemical has been classified, the hazard(s) must be communicated to target audiences

Labels and Safety Data Sheets are the main tools for chemical hazard communication

Label Elements

Product Identifier

Signal Word (Danger, Warning)

Hazard Statement (for each hazard class and category) Pictogram(s)

Precautionary Statement (for each hazard class and category)

Name, Address and Tele. # of chemical manufacturers

Signal Words

"Danger" or "Warning" are used to emphasize hazards and indicate the relative level of severity of the hazard, assigned to a hazard class and category.

"Danger" for the more severe hazards

"Warning" for the less severe hazards.

Hazard Statement

Hazard statements are standardized and assigned phrases that describe the hazard(s) as determined by hazard classification

“Causes eye irritation”

“Toxic if inhaled”

“Flammable aerosol”

“May cause cancer”

“May cause respiratory irritation”

“Harmful to aquatic life”

“Contains gas under pressure; may explode if heated”

Symbols/Pictograms

Convey health, physical and environmental hazard information, assigned to the hazard class and category.

The symbols have been incorporated into pictograms for use on the label

Pictograms will have a black symbol on a white background with a red diamond frame

Health Hazard



Carcinogen
Mutagenicity
Reproductive Toxicity
Respiratory Sensitizer
Target Organ Toxicity
Aspiration Toxicity

Environment



Aquatic toxicity

Exclamation Mark



Irritant (skin and eye)

Skin Sensitizer

Acute Toxicity

Narcotic Effects

Respiratory Tract Irritant

Hazardous to Ozone Layer (Non-Mandatory)

Skull & Crossbones

Acute toxicity (fatal or toxic)



Gas Cylinder

Gases Under Pressure



Corrosion



Skin Corrosion/Burns
Eye Damage
Corrosive to Metals

Exploding Bomb

Explosives
Self-Reactives
Organic Peroxides



Flame Over Circle

Oxidizers



Flame



Flammables
Pyrophorics
Self-Heating
Emits Flammable Gas
Self-Reactives
Organic Peroxides

Physical Hazard Pictograms



Health Hazard Pictograms



Unlabelled Chemicals

However, if the chemical is unlabelled, there are some simple tests you can perform to help you identify whether it's hazardous. You might, for example, carry out a **flame test** using a small sample, or you may want to do a smell test. There are also more advanced laboratory instruments that you can use, such as **mass spectrometers**. When in doubt, always take all the necessary precautions.

How can you reduce the risk of hazardous chemicals?

Laboratories and chemical factories have protocols on how to reduce the risk of hazardous chemicals. Workers are required **to wear protective clothing, goggles, and gloves**. Hazardous chemicals should also be stored in accordance with the relevant guidelines, such as at the appropriate temperature and pressure.

Minimizing hazardous chemical exposure

Minimizing your exposure to hazardous chemicals is relatively straightforward. For starters, never handle a chemical that isn't part of your tasks at work. If you do need to deal with hazardous substances, it's essential to wear the **appropriate PPE** and follow the correct protocols in handling and disposing of such chemicals. You should also ensure that the work area is properly ventilated.

Atom economy



Barry M. Trost
Stanford University

Atom-Economy is a calculation based on the overall balanced chemical equation. It is simply the mass of desired product divided by the total mass of products.

Or, since the mass of products equals the mass of reactants in a balanced chemical equation, atom economy is the mass of desired product divided by the total mass of reactants.

Percentage Yield

- In industry manufacturers will want to know how efficient their process is.
- This can be done by using the following equation:

$$\text{Percentage Yield} = \frac{\text{Actual Yield}}{\text{Theoretical Yield}} \times 100$$

Theoretical Yield: Mass of product expected from a chemical reaction

Actual Yield: Mass of product actually obtained from carrying out the reaction

Percentage Yield

What is missing?

- ☐ What co-products are made?
- ☐ How much waste is generated?
- ☐ Is the waste benign waste?
- ☐ How much energy is required?
- ☐ Are purification steps needed?
- ☐ What solvents are used? (are they benign and/or reusable?)
- ☐ Is the “catalyst” truly a catalyst? (stoichiometric vs. catalytic)

Green chemistry & Industrial Ecology

ESTABLISHED GREEN CHEMISTRY METRICS

1. Atom economy (atom efficiency)

$$\text{Atom Economy} = \frac{\text{Mass of desired product*}}{\text{Mass of all reactants*}}$$

**including the stoichiometric coefficient*

2. Environmental (E) factor

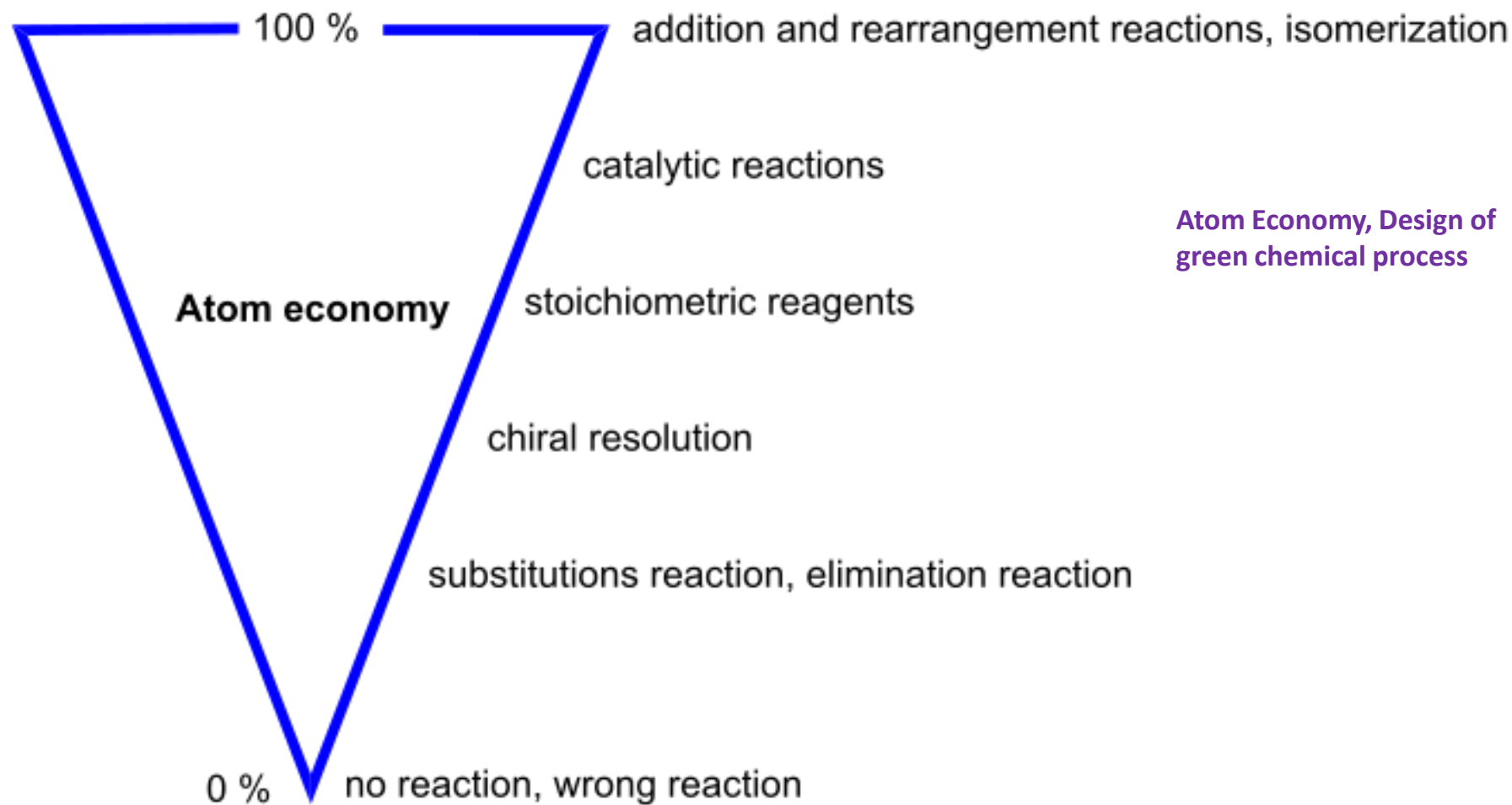
$$\text{E-factor} = \text{total waste (g)} / \text{product (g)}$$

3. Atom utilization (variation of atom economy)

$$\% \text{ Atom Utilization} = (\text{MW of desired product} / \text{MW of all products}) \times 100$$

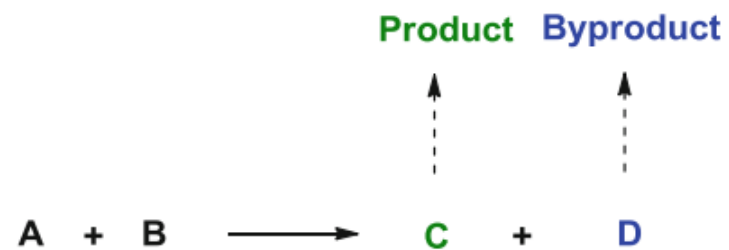
4. Reaction mass efficiency (RME)

$$(\text{mass of product C} / (\text{mass of A} + \text{mass of B})) \times 100$$



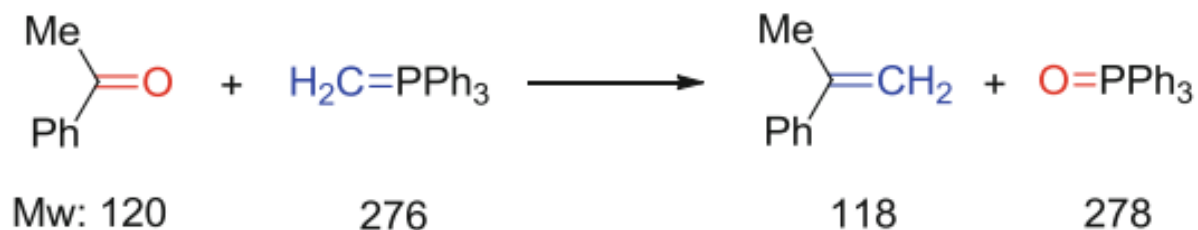
Atom Economy, Design of
green chemical process

1) Traditional reaction

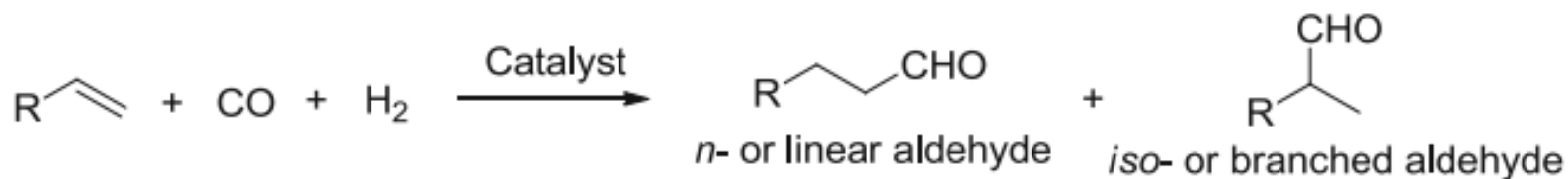


2) Atom economic reaction



Case 1

$$\text{Atom economy (\%)} = \frac{118}{120 + 276} \times 100\% = 29.79\%$$

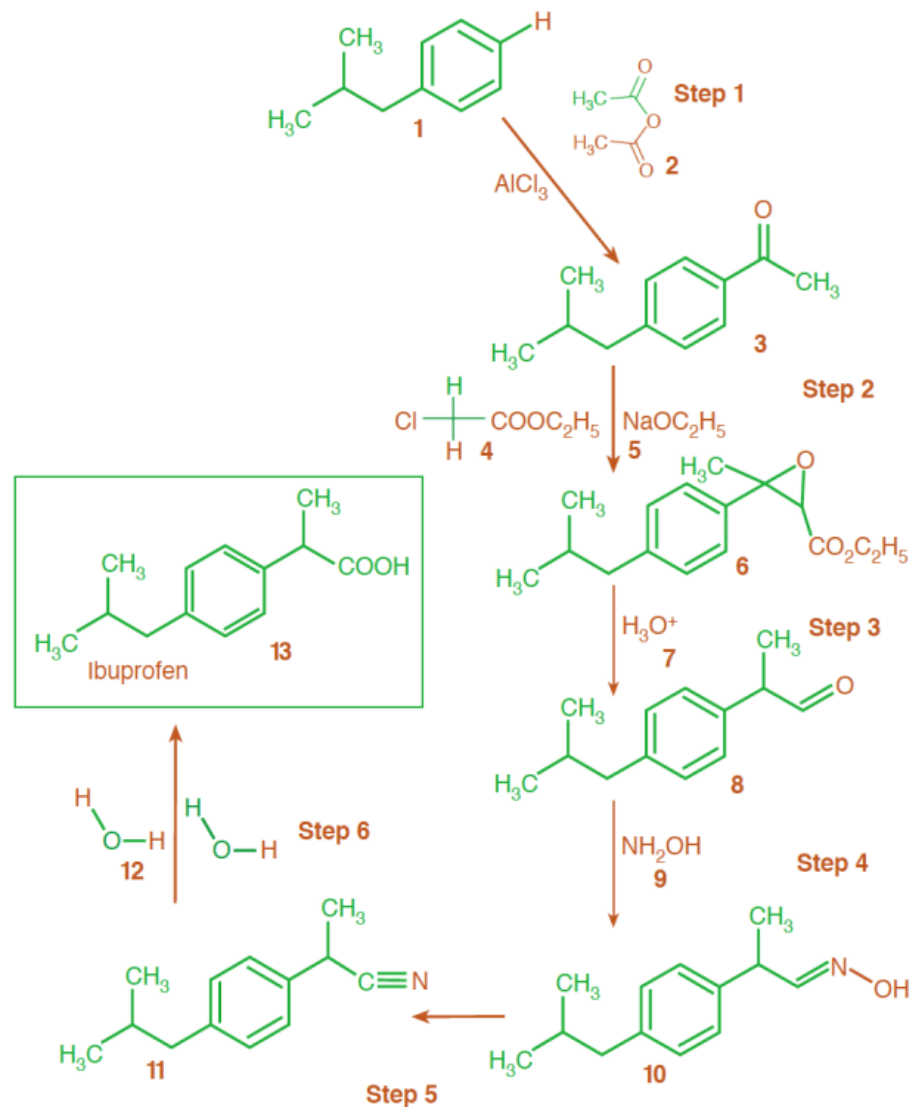
Case 2

$$\text{Atom economy (\%)} = 100\%$$

Atom economy example: Ibuprofen synthesis

The Problem:

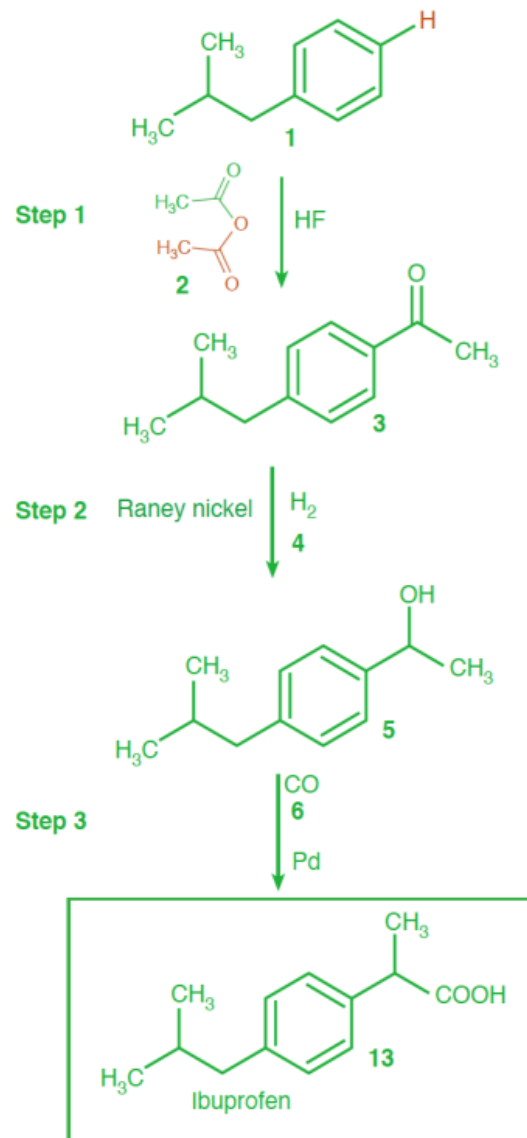
The traditional industrial synthesis of ibuprofen was developed and patented by the Boots Company of England in the 1960s (U.S. Patent 3,385,886). This synthesis is **a six-step process** and **results in large quantities of unwanted waste** chemical byproducts that must be disposed of or otherwise managed. Much of the waste that is generated is a result of **many of the atoms of the reactants not being incorporated into the desired product** (ibuprofen) but into unwanted byproducts (poor atom economy/atom utilization). The process also uses a variety of solvents, and catalysts (if any) are not used in stoichiometric amounts.



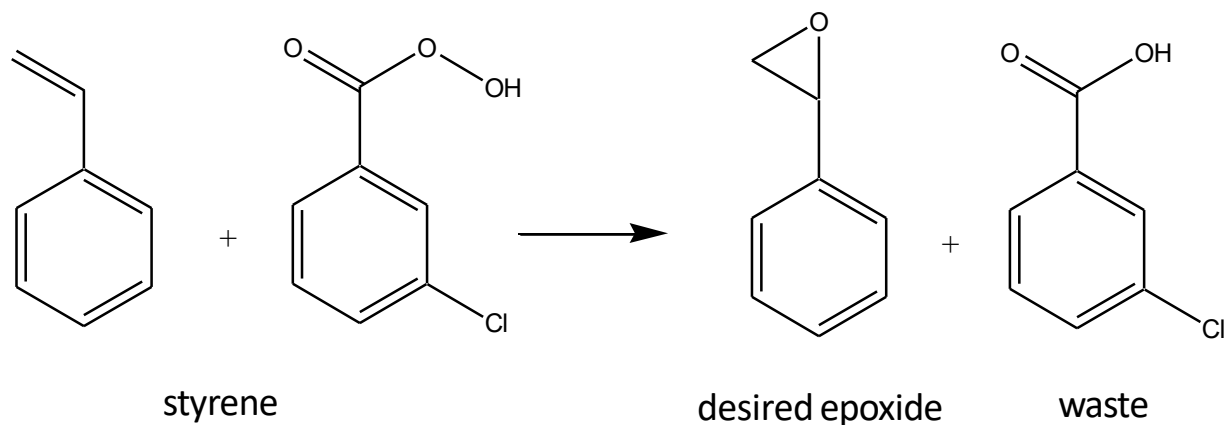
Atom economy example: Ibuprofen synthesis

The Solution:

The BHC Company has developed and implemented a new greener industrial **synthesis of ibuprofen that is only three steps** (U.S. Patents 4,981,995 and 5,068,448, both issued in 1991). In this process, **most of the atoms of the reactants are incorporated into the desired product** (ibuprofen). This results in only small amounts of unwanted byproducts (very good atom economy/atom utilization) thus lessening the need for disposal and mediation of waste products. There are other environmental advantages to the green synthesis versus the brown synthesis, including solvents and catalyst.



Example: Epoxidation of styrene



- Assume 100% yield.
- 100% of the desired epoxide product is recovered.
- 100% formation of the co-product: m-chlorobenzoic acid.
- A.E. of this reaction is 23%.
- 77% of the products are waste.

Case study: Phenols

Phenols are widely used in household products and as intermediates for industrial synthesis. They are often referred to as drop-in platform chemicals.

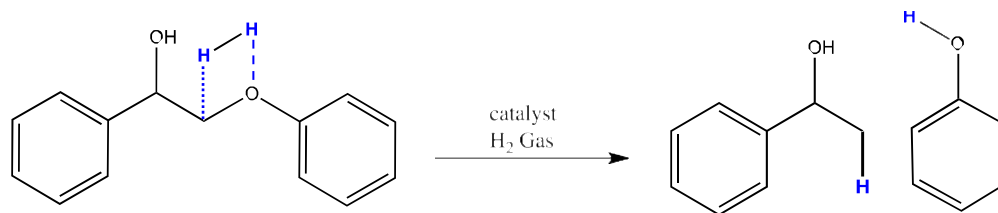
**Conventional method of
obtaining phenols from
petroleum:**

- Petroleum
- The method is not sustainable –
it is
dependent on depleting
resources.

Case study: Phenols

Alternative production of phenols from biomass waste using depolymerization:

- This method uses abundant product (waste) as a starting material.



Biomass

Phenols

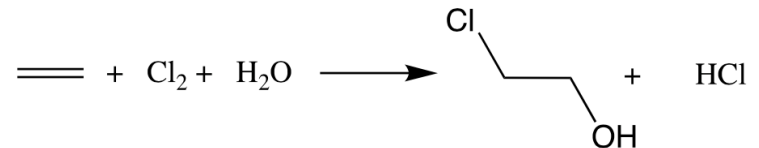
Case Study: Production of Ethylene Oxide

Ethylene oxide is used as an intermediate in the production of several industrial chemicals, the most notable of which is ethylene glycol. It is also used as a fumigant in certain agricultural products and as a sterilant for medical equipment and supplies.

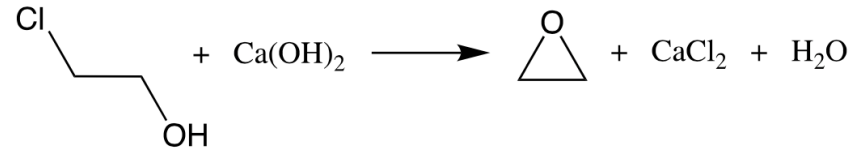
Conventional ethylene oxide synthesis included:

- A 2-step synthesis with a chlorohydrin intermediate.
- For each kilogram of product, 5 Kg of waste were disposed.

Step 1:



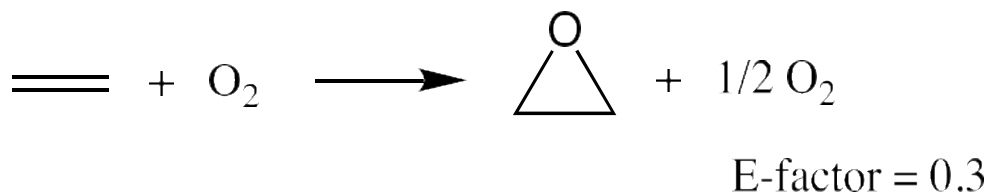
Step 2:



Case Study: Production of Ethylene Oxide

Alternative production of ethylene oxide:

- Use of molecular oxygen removes the need for chlorine.
- New process generates more than 16 times less waste than the original one, and eliminates the formation of waste water.



Safer Solvents and Auxiliaries. The use of auxiliary substances (solvents, separation agents, etc.) should be made unnecessary whenever possible and, when used, innocuous.

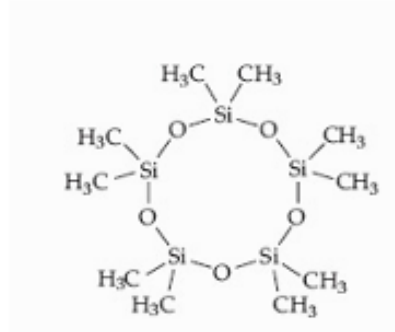
“CO2, when compressed into a liquid state, has long been recognized as an ideal solvent.

CO2 solutions are nontoxic, nonflammable, energy-efficient, cost-effective, waste minimizing, and reusable.

Through green chemistry initiatives, CO2 has replaced many polluting solvents.”

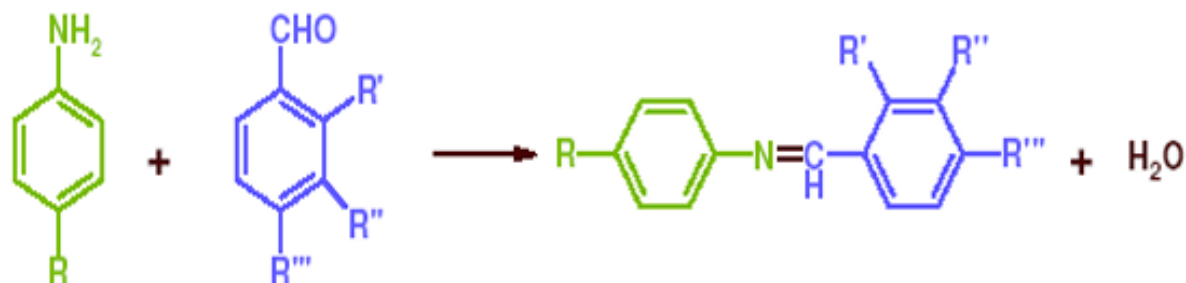
Preferred	Useable	Undesirable
Water	Cyclohexane	Pentane
Acetone	Heptane	Hexane(s)
Ethanol	Toluene	Di-isopropyl ether
2-Propanol	Methylcyclohexane	Diethyl ether
1-Propanol	Methyl t-butyl ether	Dichloromethane
Ethyl acetate	Isooctane	Dichloroethane
Isopropyl acetate	Acetonitrile	Chloroform
Methanol	2-MethylTHF	Dimethyl formamide
Methyl ethyl ketone	Tetrahydrofuran	N-Methylpyrrolidinone
1-Butanol	Xylenes	Pyridine
t-Butanol	Dimethyl sulfoxide	Dimethyl acetate
	Acetic acid	Dioxane
	Ethylene glycol	

“Dry Cleaning of Clothes: perchloroethylene versus Liq. CO2



Undesirable Solvent	Alternative
Pentane	Heptane
Hexane(s)	Heptane
Di-isopropyl ether or diethyl ether	2-MeTHF or <i>tert</i> -butyl methyl ether
Dioxane or dimethoxyethane	2-MeTHF or <i>tert</i> -butyl methyl ether
Chloroform, dichloroethane or carbon tetrachloride	Dichloromethane
Dimethyl formamide, dimethyl acetamide or N-methylpyrrolidinone	Acetonitrile
Pyridine	Et ₃ N (if pyridine is used as a base)
Dichloromethane (extractions)	EtOAc, MTBE, toluene, 2-MeTHF
Dichloromethane (chromatography)	EtOAc/heptane
Benzene	Toluene

A solventless reaction:



Less hazardous chemical synthesis

Organic chemistry's synthetic toolbox has been significantly improved over the past decade by the innovative work of green chemists and the many new reactions that have been developed.

Cascade, or tandem reactions, such as C–H activation, metathesis, and enzymatic reactions are some excellent examples of the cleaner and more efficient synthetic tools that are now available to organic chemists.

Cascade, or tandem reactions

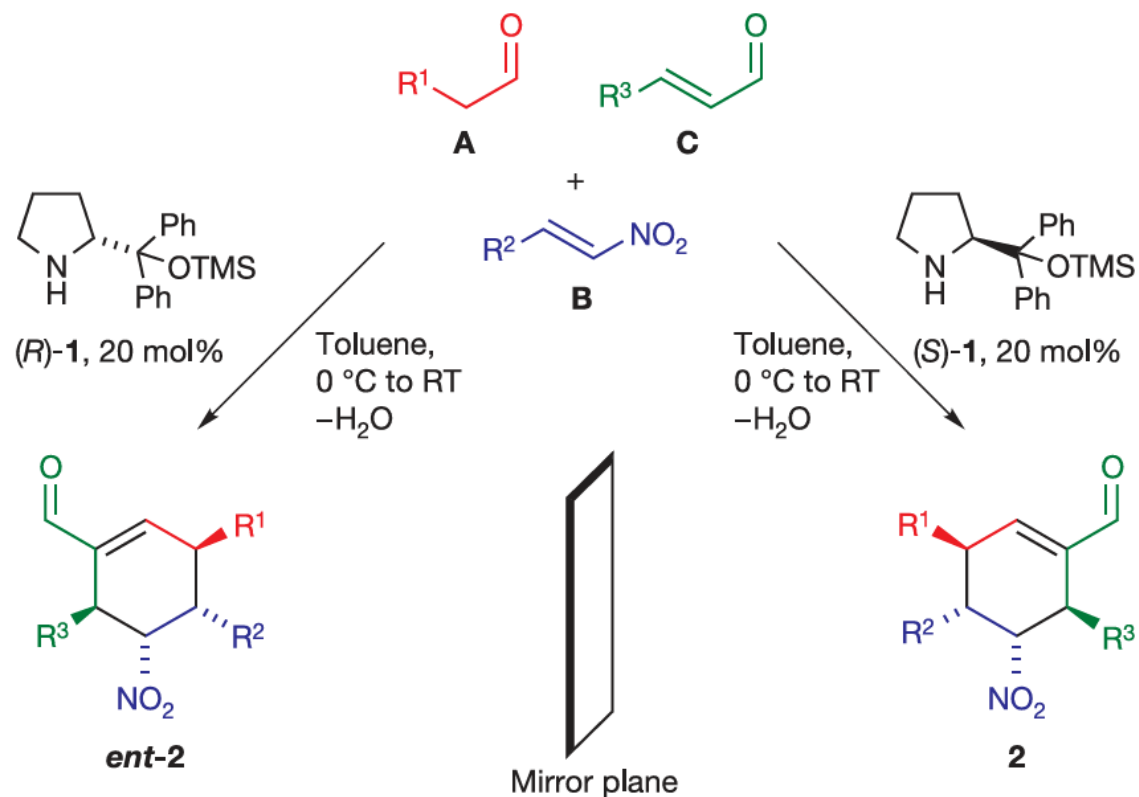
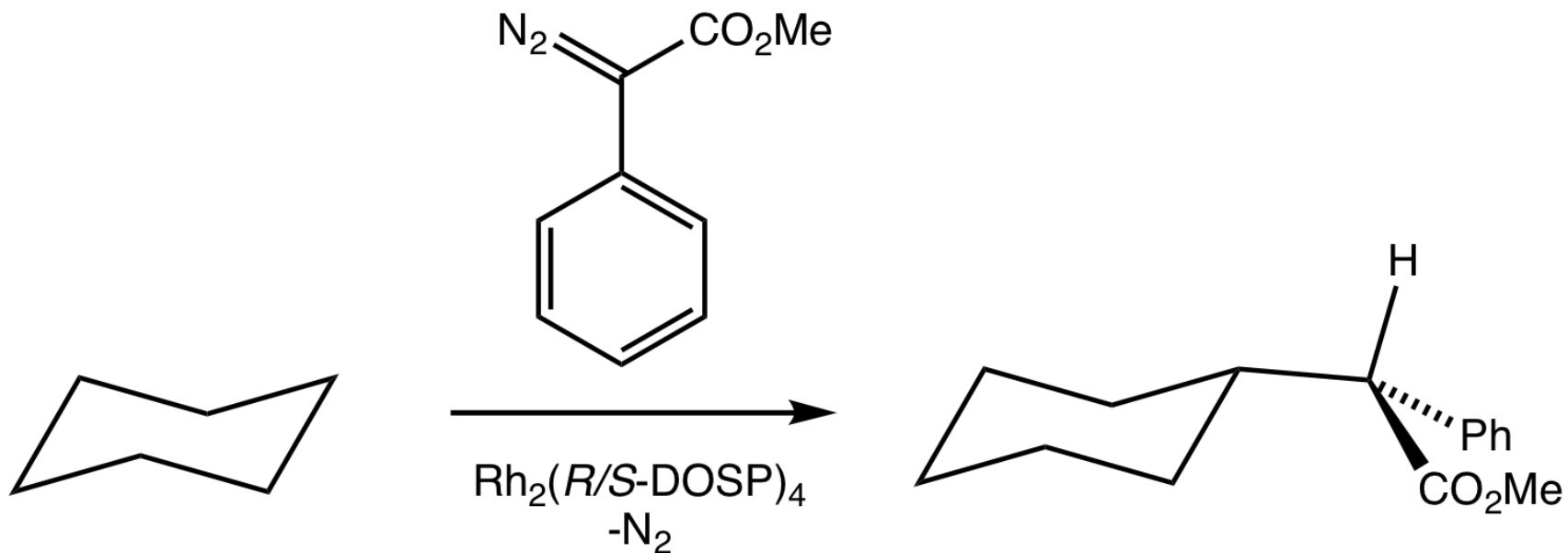


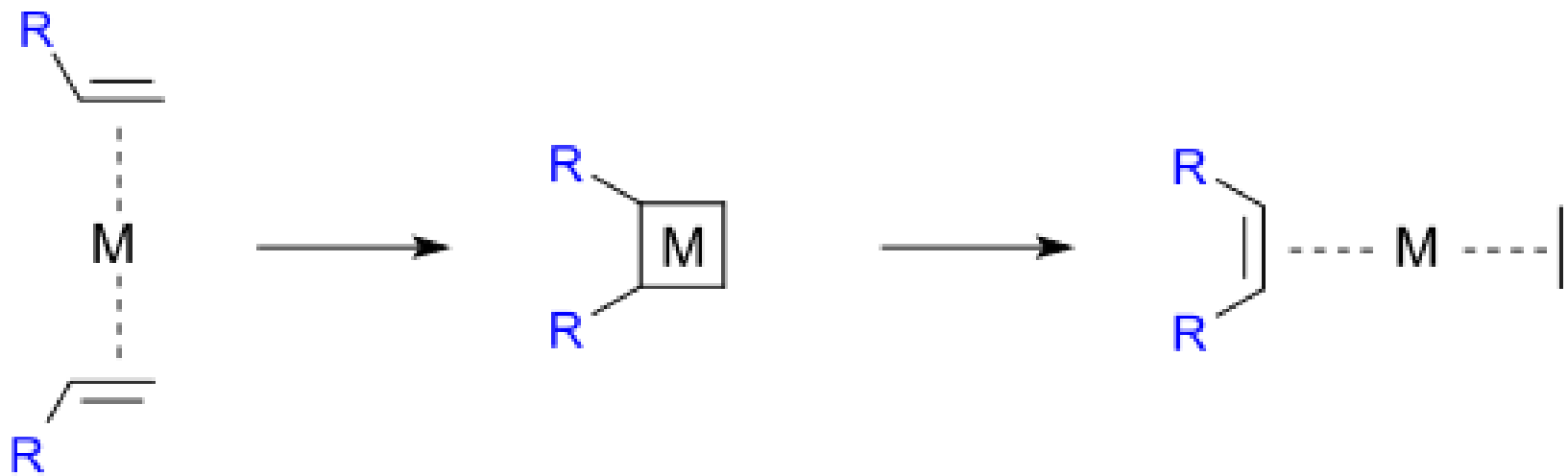
Figure 1 | Asymmetric, organocatalytic three component multistep reaction cascade. RT, room temperature.

C–H activation

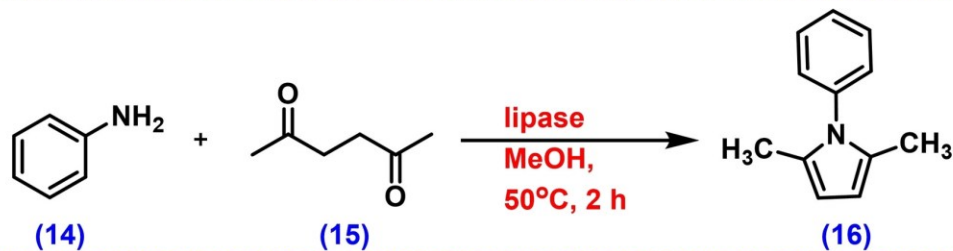


Davies, H. M. L.; Hansen, T. *J. Am. Chem. Soc.* **1997**, 119, 9075.

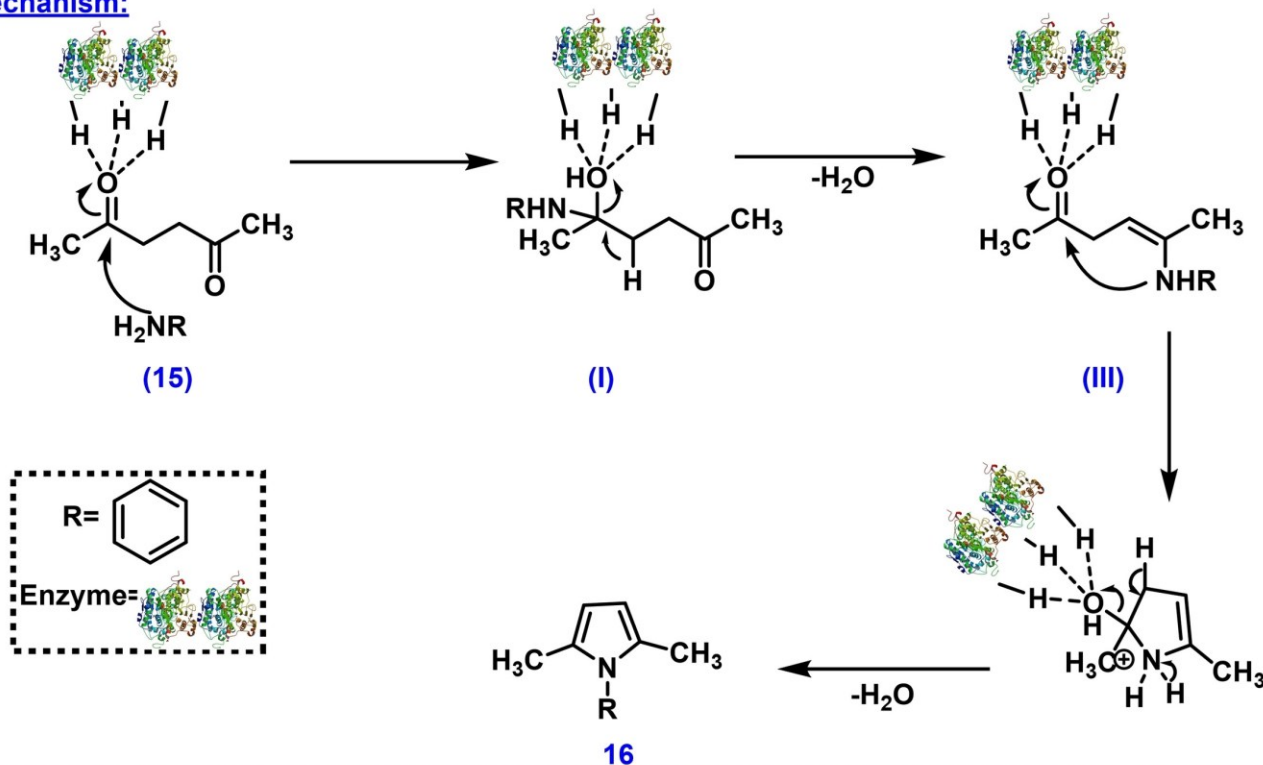
Metathesis



Enzymatic reactions

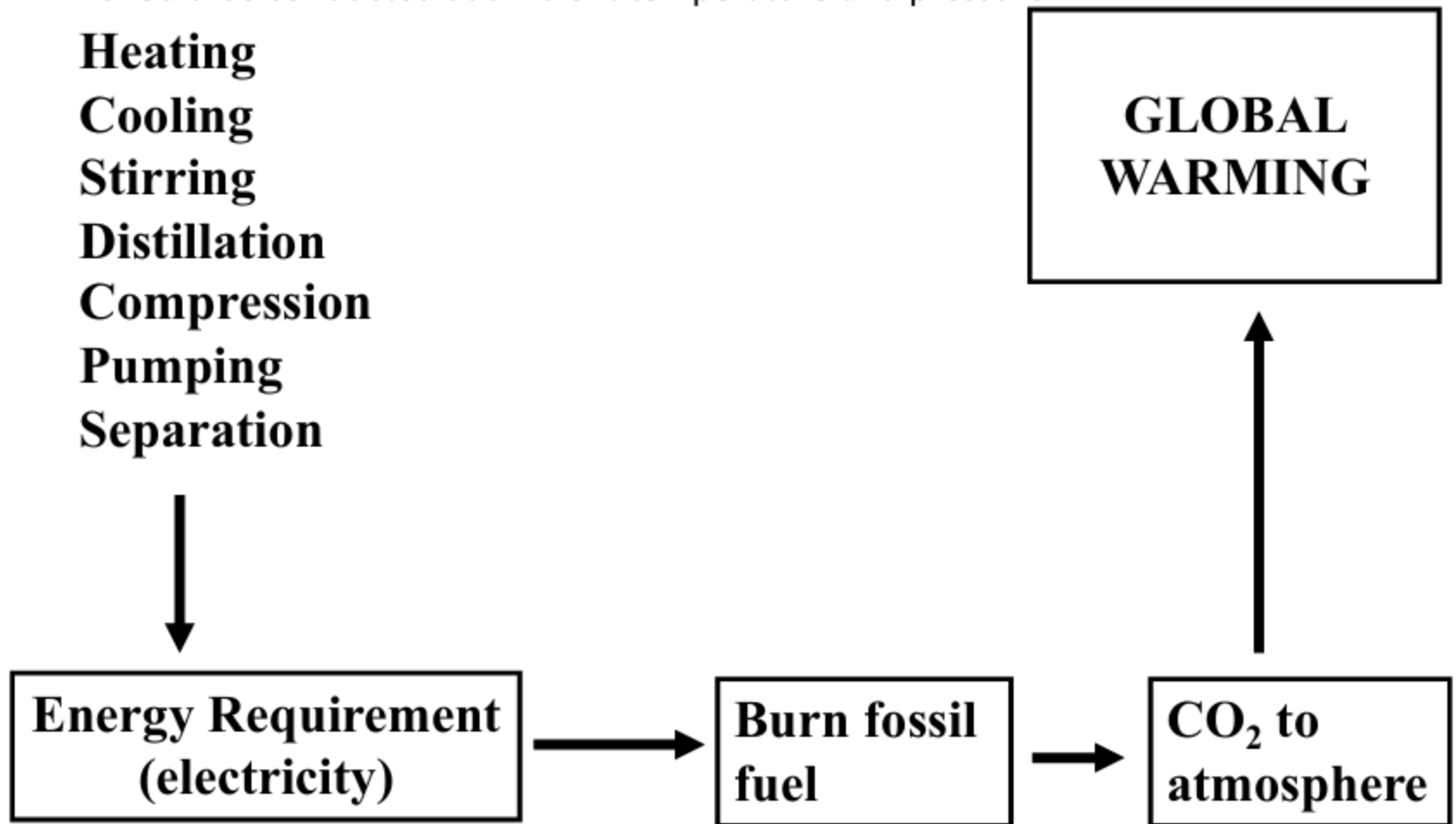


Mechanism:

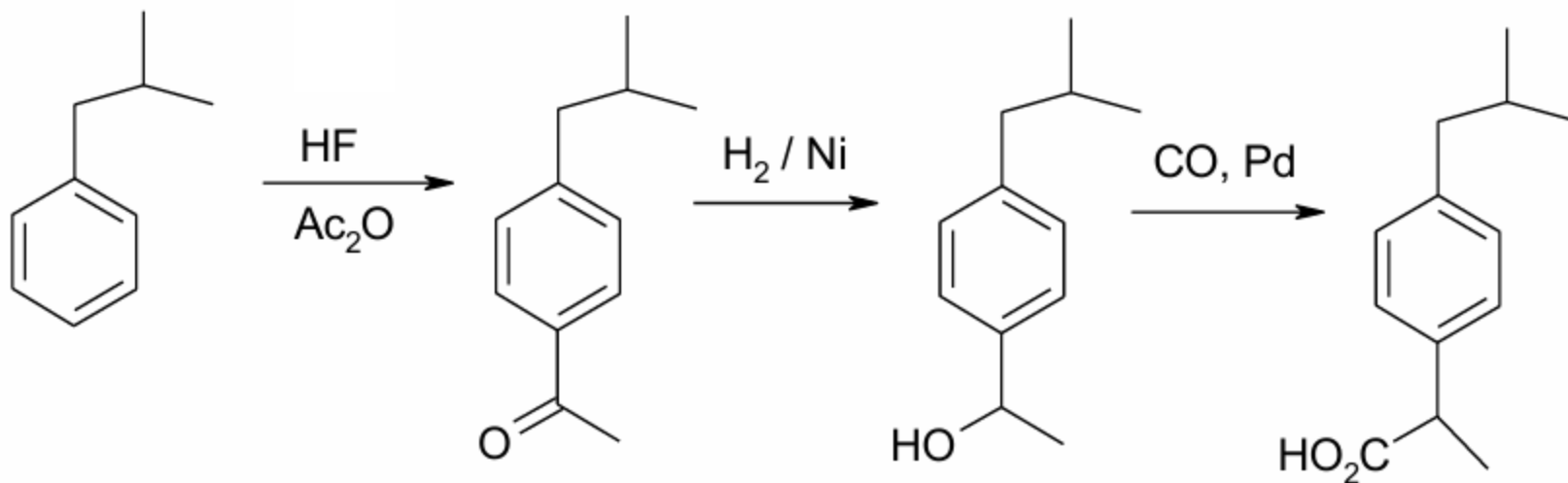


Lipase catalyzed synthesis of *N*-substituted pyrrole.

Design for Energy Efficiency. Energy requirements should be recognized for their environmental and economic impacts and should be minimized. Synthetic methods should be conducted at ambient temperature and pressure.

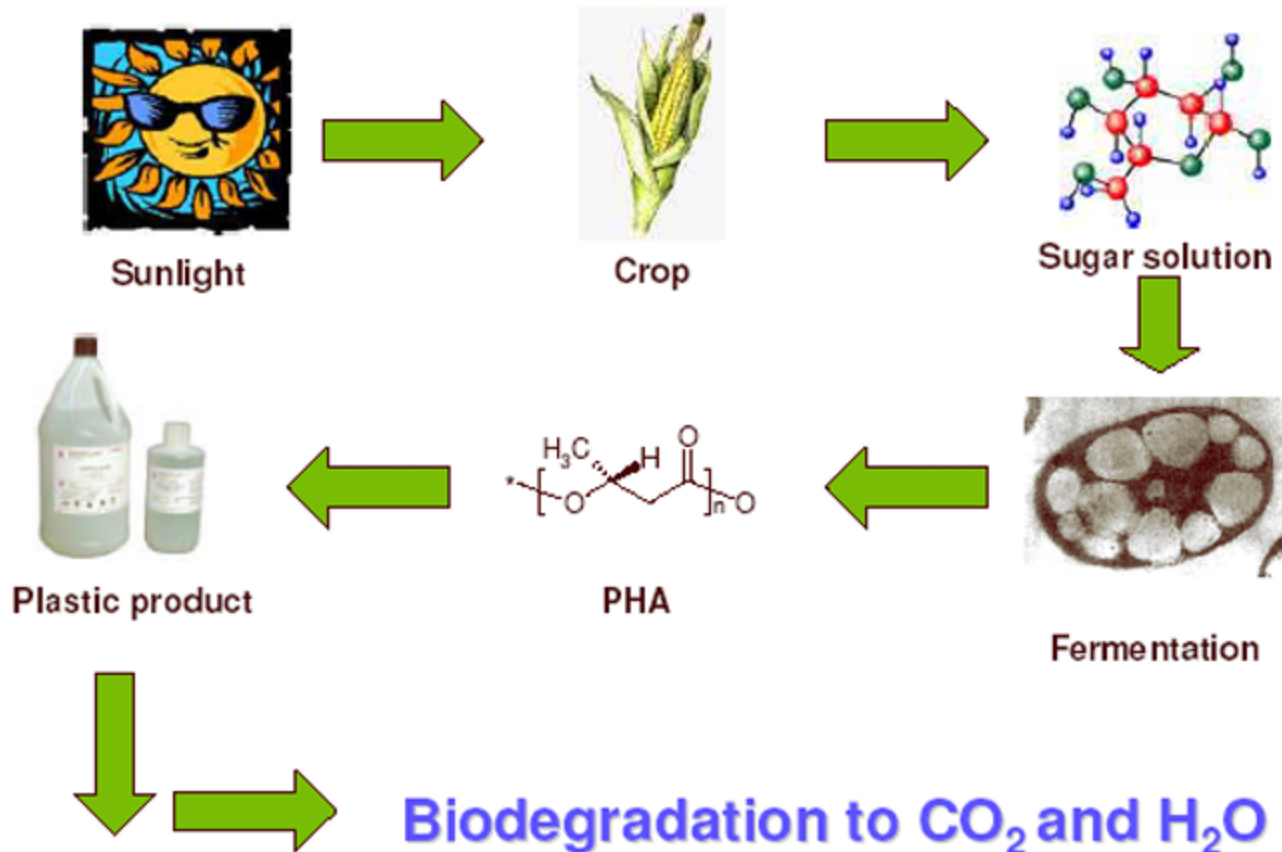


Catalysis. Catalytic reagents (as selective as possible) are superior to stoichiometric reagents



Design for Degradation. Chemical products should be designed so that at the end of their function they do not persist in the environment and instead break down into innocuous degradation products

Polyhydroxyalkanoates (PHA's)



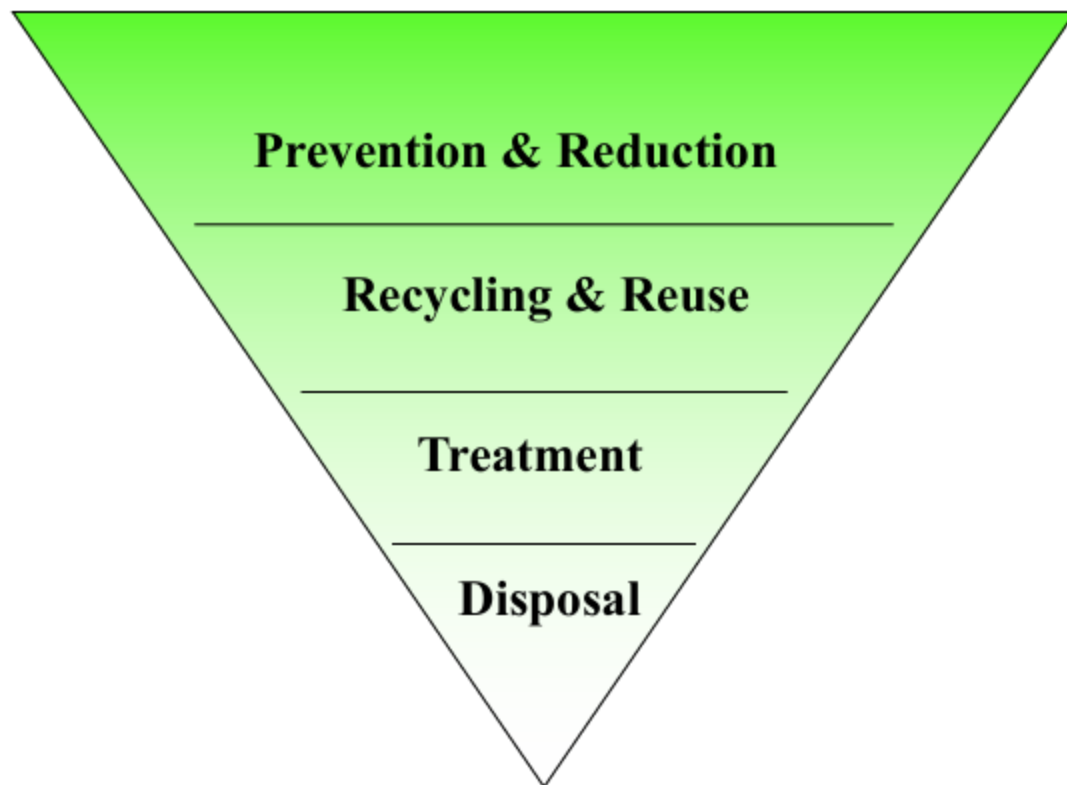
Real-time Analysis for Pollution Prevention. Analytical methodologies need to be further developed to allow for real-time in-process monitoring and control prior to the formation of hazardous substances.

Inherently Safer Chemistry for Accident Prevention. Substance and the form of a substance used in a chemical process should be chosen so as to minimize the potential for chemical accidents, including releases, explosions, and fires.

Tragedy in Bhopal, India - 1984

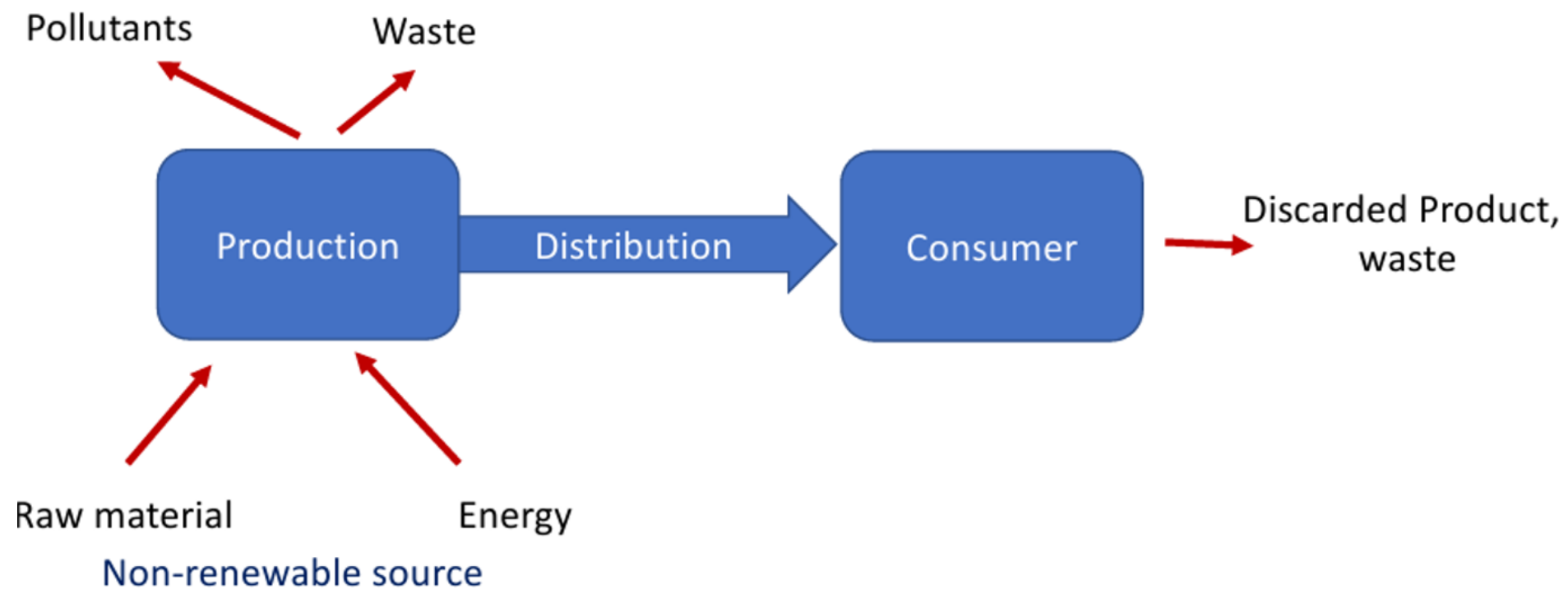
In arguably the worst industrial accident in history, 40 tons of **methyl isocyanate** were accidentally released when a holding tank overheated at a Union Carbide pesticide plant, located in the heart of the city of Bhopal. 15,000 people died and hundreds of thousands more were injured.

Pollution Prevention Hierarchy



Green chemistry **Not** a solution to all environmental problems **But** the most fundamental approach to preventing pollution.

Industrial ecology



Industrial ecology

integrates the principles of science, engineering, and ecology in industrial systems

through which goods and services are provided in a way that minimizes environmental impact and optimizes utilization of resources, energy, and capital

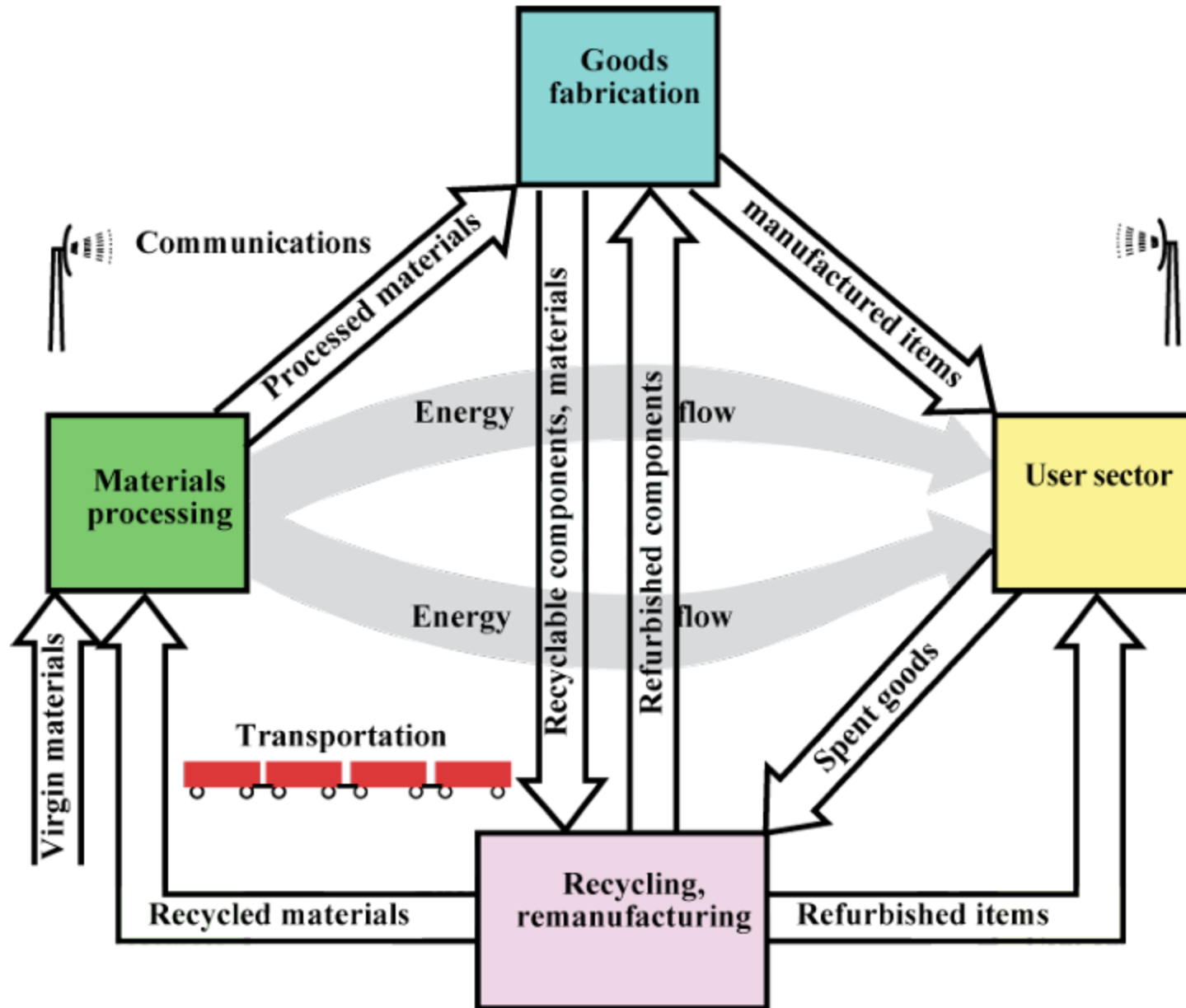
Every aspect of the provision of goods and services from concept, through production, and to the final fate of products remaining after use

$$\frac{\text{Market value of products}}{\text{Consumption of material and energy}}$$

An **industrial ecosystem** functions through groups of industrial concerns, distributors, and other enterprises functioning to mutual advantage, using

each others' products,
recycling each others' potential waste materials,
and utilizing energy as efficiently as possible to maximize

Major Components of an Industrial Ecosystem Showing Maximum Flows of Material and Energy Within the System



Industrial Ecology:

- ❖ Saving resources
- ❖ Recycling
- ❖ Be efficient when possible

Why?

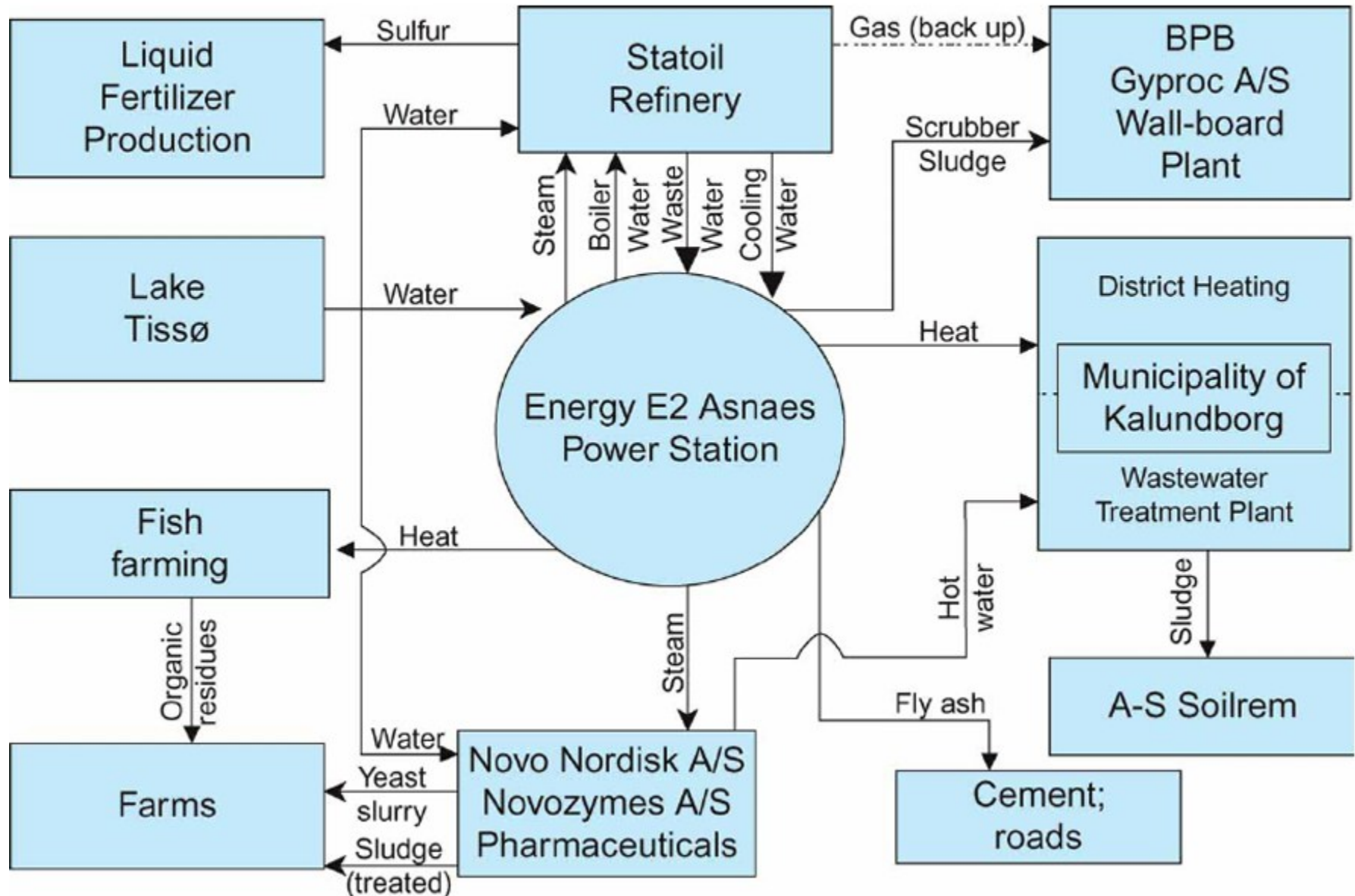
Fewer resources consumed
→ lower operational costs

Less waste/trash
→ lower disposal costs

- Material, water, and energy flows
- Companies within close proximity
- Strong informal ties between plant managers
- Minor retrofitting of existing infrastructure
- One or more anchor tenants.

The most often cited example of a functional industrial ecosystem : Kalundborg, Denmark

The Kalundborg Industrial Ecosystem



Saves resources:

- 30% better utilization of fuel using combined heat + power than producing separate
- Reduced oil consumption
- less oil-burning heaters in homes
- Does not drain fresh water supplies
- New source of raw materials
- Gypsum, sulfuric acid, fertilizer, fish farm

1. Which of the below statements is/are correct?

- a) Petrol mainly contains hydrocarbons in the C_8-C_{15} range.
- b) Kerosene contains hydrocarbons roughly in the $C_{10}-C_{16}$ range.
- c) Jet fuel is chemically similar to kerosene.
- d) LPG is similar to Jet fuel

2. The principle of "Atom Economy" focuses on:

- a) Increasing the selectivity of byproducts
- b) Maximizing the incorporation of reactants into the final product
- c) Enhancing the recyclability of solvents
- d) Reducing energy consumption

